Aleksei Nemtyrev

Statistical Modelling · Causal Inference · Data Science

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Profile

Quantitative researcher finishing a PhD in econometrics (2027). My work is about separating signal from noise: I design estimators that cut variance without losing what matters, and evaluate policy rules that could make the economy less volatile. I build them in Python, R and MATLAB and teach Data Science Methods at MSc level. Before my PhD, I ran a retail company, taking it from break-even to profitability within a year.

Experience

PhD Candidate Aug 2023 – 2027 (expected)
Tilburg University, Econometrics & Operations Research Tilburg, NL

- Build estimators end to end: from mathematical proof to large-scale simulation to a public replication package.
- Run simulations on an HPC cluster (MATLAB Parallel Computing Toolbox): 20M+ bootstrap replications per study.
- Develop modern macro (DSGE) models and take them to data, from monetary policy rules to model testing.
- Talks at IAAE 2026, CFE-CMStatistics 2026, GIFT 2026, NESG 2024 & 2026, Econometric Society Winter Meeting 2025; co-authors at NBER-NSF 2026, $(EC)^2$ 2026, Chicago Fed 2025, Purdue 2024, Erasmus 2024.
- Teach *Data Science Methods* to 120 MSc students a year (in R: PCA, ridge/lasso, trees, double ML), with my own model-selection and PCA assignments; also *Statistics 2* to 300 BSc students a year.

General Manager Jun 2016 – Jul 2020
NEVA Torg (retail) St. Petersburg, RU

- Full operational and financial responsibility (accounting, personnel, suppliers); managed a team of 10.
- Grew the chain to seven stores at its peak (opened three, closed one); revenue per store rose 30% in 2018.
- Took the company from break-even in 2017 to RUB 2.5m (≈€35k) after-tax profit in 2018, a net margin of ∼9%.

Selected Projects

Targeted Local Projections with O. Boldea · working paper, 2026
Combines two estimators, local projections and VARs, to optimise the bias–variance trade-off at every horizon. paper code

FOMC Responses to Credit Spreads with J. R. Campbell · in progress
Proposes adding credit spreads to the Fed's policy rule: could cut consumption and unemployment volatility by 20% and 30%.

Corrected Local Projections for Structural IRFs with O. Boldea, A. Cornea-Madeira, A. R. Hall · in progress
Estimator that strips predictable noise out of local projections to reduce their variance.

DSGE Identification and Testing with J. R. Campbell · in progress
Novel utility function and model simplification make a macro model estimable and testable directly from data.

Education

PhD, Econometrics, *Tilburg University* 2023 – 2027 (expected)
Research Master, Economics, *Tilburg University · CentER Scholarship* 2021 – 2023
MSc Economics, Monetary Policy & Banking, *University of Amsterdam · Amsterdam Merit Scholarship* 2020 – 2021
MA (Hons) Economics, *University of Glasgow* 2012 – 2016

Skills

Machine learning	Bias–variance trade-off, ridge, lasso, cross-validation, PCA, classification, random forests, boosting
Causal inference	Difference-in-differences, IV, GMM, double ML, structural VARs, local projections
Time series	VARs, ARMA models, state-space models, Kalman filter, impulse-response analysis
Statistics	Bayesian inference (MCMC), maximum likelihood, bootstrap, Monte Carlo simulation
Programming	Python (MCMC sampler, climate–macro model), R, MATLAB, Stata, LaTeX, Git
AI workflows	Agentic AI (Claude Code, Codex): research codebases, simulation pipelines, document automation
Domain	Macro-finance, monetary policy, credit spreads, FRED and Fed data, DSGE models
Business	General management, team leadership, accounting, supplier management, retail operations
Languages	English (C2), Russian (C2), Dutch (B1)